SOC Flagship Investigation Report

Cross platform lab analysis in Microsoft Sentinel

Evidence aligned revision | 14 October 2026

Author: William Gokah | WilliamInCyber
Workspace: law-soc-lab, Azure East US
Platform: Microsoft Sentinel in the unified Defender portal

1. Executive summary

This report consolidates two authorized home lab exercises: SSH password guessing against an Ubuntu account and benign PowerShell execution tests on JAMES-VM. It describes the observed records, collection configuration, query logic and documented response steps. These are simulated activities, not production incidents.

The Linux evidence shows 88 failed password records grouped under source 192.168.64.15 and account mary. A separate correlation on source IP alone returns 88 failures and eight accepted password records. The detailed view includes accepted password activity for mary. A UFW source deny was configured, but a connection test after the change is not pictured.

The Windows evidence shows a benign encoded command test, an execution policy bypass test and a PowerShell event 4104 hunt returning eight records. The captured EventData values are collapsed or truncated. Complete decoded payload recovery, detection of every simulated variant and incident closure are not established by this evidence set.

2. Environment and collection

Component	Documented scope
Linux	Ubuntu host; Computer field wazuh-manager; SSH account mary
Windows	JAMES-VM; Windows PowerShell Operational event 4104
Collection	Project notes document Azure Arc, Azure Monitor Agent and DCR configuration
Destination	Linux records in Syslog; captured Windows hunt records in Event
Lab source	192.168.64.15; Hydra exercise documented in the Day 5 write-up

The Syslog baseline sample includes authpriv and kern facilities. It does not establish that the entire table contains only authentication telemetry. The Windows DCR view selects Security event 4688 and PowerShell Operational event 4104. Successful configuration commands and destination query results are separate evidence stages.

An empty SecurityEvent baseline over Last 24 hours is a scoped observation, not proof of tenant gating. Microsoft documents that general Windows Event Logs DCR collection uses Event; the Sentinel Windows Security Events via AMA connector uses SecurityEvent. The screenshots do not establish the precise cause of the empty baseline.

3. Linux authentication investigation

Context: Controlled Hydra SSH password guessing exercise against mary. **Source:** 192.168.64.15. **Query window:** Last 24 hours in the captured summaries.

Failure grouping: captured query

```
Syslog
| where SyslogMessage has "Failed password"
| extend SourceIP = extract(@"from (\d+\.\d+\.\d+\.\d+)", 1, SyslogMessage)
| extend TargetUser = extract(@"for (\w+)", 1, SyslogMessage)
| summarize FailedAttempts = count() by SourceIP, TargetUser
| sort by FailedAttempts desc
```

The result shows 88 records for the identified source and account, plus two records with blank extracted fields. The two unparsed records are not assigned to the source or account without their original messages. The extraction is specific to the message patterns it matches and is not a complete parser for every SSH authentication format.

Failure and success correlation: captured query

```
let failures = Syslog
    | where SyslogMessage has "Failed password"
    | extend SourceIP = extract(@"from (\d+\.\d+\.\d+\.\d+)", 1, SyslogMessage)
    | summarize Failures = count() by SourceIP;
let successes = Syslog
    | where SyslogMessage has "Accepted password"
    | extend SourceIP = extract(@"from (\d+\.\d+\.\d+\.\d+)", 1, SyslogMessage)
    | summarize Successes = count() by SourceIP;
failures
| join kind=inner successes on SourceIP
| where Failures > 10 and Successes > 0
| project SourceIP, Failures, Successes
```

This returns 88 failures and eight successes for one source. It does not retain account or destination host, or establish temporal order. The threshold is a lab query condition, not a production threshold validated against a baseline. The detailed source filtered view returns 96 records and includes failed and accepted password activity for mary; complete times and messages are needed for precise sequencing. That total equals 88 failures plus eight successes, so the two unparsed failure messages do not contain this source IP.

Documented response and disposition

```
sudo ufw insert 1 deny from 192.168.64.15
sudo ufw status numbered
```

The terminal shows active UFW, a source deny at rule 1 and the general SSH allow at rule 3. The deny is broader than SSH alone. This establishes the configuration, not a verified block of new connections, termination of an existing session, credential reset or completed recovery. Disposition: lab activity investigated and a response configuration documented; response effectiveness remains unverified in the reviewed screenshots.

4. Windows PowerShell investigation

Context: Benign execution tests on JAMES-VM. **Observed collection:** Event table, EventID 4104.
Status: Candidate script block records located; full content correlation remains an evidence checkpoint.

Known test actions

The encoded test converts a Write-Output marker to Base64 using Unicode encoding, then passes it to powershell.exe with -NoProfile and -EncodedCommand. The console prints SOC-DAY6-ATTACK-ENCODED-PAYLOAD. This shows the submitted test and its console output; a matching expanded event is needed to demonstrate payload recovery in the workspace.

```
powershell.exe -NoProfile -ExecutionPolicy Bypass -Command
    "Write-Output 'SOC-DAY6-ATTACK-BYPASS'"
```

The bypass command above is wrapped across lines for display; the screenshot executes it on one line and prints SOC-DAY6-ATTACK-BYPASS. The file named day06-powershell-attack-cradle.png shows this bypass test. Its filename is not evidence of a download cradle. No transfer outcome is established by that image.

Hunt query: aligned with the Part 6 capture

```
Event
| where EventID == 4104
| where TimeGenerated > ago(30m)
| where EventData has_any (
    "DownloadString", "IEX", "Invoke-Expression",
    "FromBase64String", "Net.WebClient", "SOC-DAY6-ATTACK"
)
| project TimeGenerated, Computer, EventData
| sort by TimeGenerated asc
```

The screenshot returns eight matching records, with JAMES-VM visible in the displayed rows. This revision uses the captured 30 minute filter and keyword list; Bypass is not a term in this captured run. Keyword matches are candidates for investigation and do not independently establish malicious activity. Eight records are not evidence of eight commands or of complete technique coverage.

Interpretation and limitations

The screenshot does not expand EventData. It therefore does not establish full decoded payload recovery or the relationship between each record and each test. Console and portal timestamps provide supporting context only until their time zones and complete event contents are checked. The portal also displays a schema loading error while the query returns records.

Successful auditpol and registry commands document logging configuration. The query result documents available records. Neither establishes a production compromise, automatic alert generation for this hunt, incident closure, or successful download activity. This report does not claim that hidden window execution was detected or that script block logging defeats every form of obfuscation.

5. Lab indicators and behavior mapping

These are exercise identifiers and investigative pivots, not a threat intelligence blocklist. A private source IP, a lab account or a PowerShell flag is not inherently malicious.

Exercise	Reviewed identifiers
Linux	192.168.64.15; account mary; OpenSSH TCP/22; host wazuh-manager
Windows	JAMES-VM; powershell.exe; EncodedCommand; ExecutionPolicy Bypass
Test markers	SOC-DAY6-ATTACK-ENCODED-PAYLOAD; SOC-DAY6-ATTACK-BYPASS

MITRE ATT&CK: exercise context

Mapping	Scope of association
T1110.001: Password Guessing	Controlled Hydra SSH password guessing documented in Day 5. T1110 is its parent technique, not an additional demonstrated test.
T1059.001: PowerShell	The captured benign encoded and bypass command tests use PowerShell. The mapping describes exercised behavior, not confirmed malicious execution.

T1105 is removed from demonstrated coverage: the reviewed screenshots do not show an ingress transfer. T1078 is not asserted as independently demonstrated adversary use of valid accounts: a join on source IP alone and accepted password records do not establish that conclusion by themselves. This mapping is not a coverage validation across all variants of either technique.

6. Capabilities documented

Collection and analysis: Workspace deployment, Azure Activity availability, Linux Syslog and Windows Event queries, field selection, filtering, extraction, grouping and correlation.

Detection configuration: Part 4 documents a custom detection with Continuous (NRT) frequency, an Enabled state and a captured lab incident. It is not described here as a scheduled analytics rule. Its displayed query does not enforce the resource group named in its description.

Investigation and reporting: Controlled Linux and Windows exercises, explicit query scope, comparison of summary and raw views, documentation of response configuration and identification of missing verification. These are portfolio lab demonstrations, not claims of professional SOC experience or production readiness.

7. Follow up verification, not completed actions

Linux: inspect the unparsed messages; correlate source, destination host, account and complete event times; test a new SSH connection after the UFW change; separately verify any session termination, credential reset and recovery steps.

Windows: expand the relevant 4104 events and preserve full script text; correlate test marker, host and time with each test; verify the 4688 collection route and query its actual destination; add genuine transfer evidence before claiming download activity or T1105 coverage.

8. Conclusion and evidence references

The seven stage project documents a Sentinel workspace build, collection checks, KQL analysis, a custom detection test and two controlled host investigations. The reviewed evidence supports the Linux source correlation, Windows script block query results and recorded configuration steps. It does not support the original report's broader claims of verified containment, complete PowerShell technique detection, fully recovered payloads or tenant gating.

This revision aligns the conclusions with the visible evidence and makes outstanding checks explicit. It revises the report's wording; it does not imply that new experiments, response tests or log expansions were performed.

Primary project evidence

Project repository
Day 5 write-up and original screenshots
Day 6 write-up and original screenshots

Evidence file	What it supports
day05-failure-spike.png	88 grouped failures; two records with blank extracted fields
day05-compromise-found.png	Correlation on source IP alone: 88 failures and eight successes
day05.b-attack-timeline.png	96 returned records; visible failed and accepted password activity
day05-containment-block.png	Active UFW and numbered source deny configuration
day06-windows-baseline-securityevent.png	Zero rows in the captured Last 24 hours baseline
day06-dcr-xpath-queries.png	Custom selections for Security 4688 and PowerShell Operational 4104
day06-logging-enabled.png	Successful auditing and logging configuration commands
day06-powershell-attack-encoded.png	Encoded benign test and printed marker
day06-powershell-attack-cradle.png	Bypass test and printed marker despite misleading filename
day06-commandline-hunt.png	Script block (4104) hunt despite command line filename; captured 30 minute query; eight records; visible JAMES-VM rows

Official technical references

Microsoft: Windows event collection and destination tables
Microsoft: about_Logging, script block logging in Windows PowerShell 5.1
MITRE ATT&CK: Password Guessing
MITRE ATT&CK: PowerShell